Data H195A : Data Science Honors Thesis Seminar

Fall 2026

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Course Location: Gateway B1015
Course Time: Tues 5pm - 7pm
Office hours: Email to find time
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https://berkeley.zoom.us/j/2034924825

Course Description

This honors seminar is the first part of a two course Honors sequence. The students will work on a full year individual original research project. Students will follow complete practices of research reproducibility and open science. The first semester will be a review of research practices and tools, and will result in the identification of a research question. The second semester will be based on carrying out an original data science project. The result will be a hybrid product with a 20-25 page (5000 - 6000 words) research paper, with data visualizations and analysis tables, along with a documented data source, annotated code, well documented GitHub repository, and open science posting of the project.

Students will work together during the class to review each other's code, documentation and reproducibility practices. Seminar time will be a combination of discussing readings, investigating approaches to research, and discussing data visualization and presentation of results. The seminar will be collaborative in discussing and reviewing best practices in Data Science.

The expectation for the Fall Semester H195A course is that students will be able to confirm their advisor and dataset during the Fall semester. Not all students will move on from H195A to H195B. Failure to make progress on the data that will be analyzed and the advisor that will work with you will result in not being able to enroll in H195B and graduate with Honors.

Learning Objectives

The specific objectives of this course are:

- Learn how to develop a research question and project
- Develop an understanding of data availability, constraints, and ethics
- Develop data management skills
- Formulate a proposal for a research project
- Prepare for Data 195B

Prerequisites

There are no specific prerequisites. However, students must be accepted into the data science honors program in order to take this course.

Academic Integrity

The student community at UC Berkeley has adopted the following Honor Code: "As a member of the UC Berkeley community, I act with honesty, integrity, and respect for others." The expectation is that students will adhere to this code. Breaches of this code will lead to penalties up to and including an F in the class and a report to the Office of Student Conduct. Students are responsible for being up to date on the academic integrity policies outlined at https://conduct.berkeley.edu/integrity/

Course Requirements

Assignment	% of Grade	Points
In-class participation	5%	5
Literature review/annotated bibliographies	5%	5
Advisor check in	5%	5
Timeline	5%	5
Methodology proposal	10%	10
Preliminary proposal presentation	10%	10
Data documentation (e.g., GitHub, MaxQDA, Box)	10%	10
Final proposal presentation	20%	20
Final proposal	30%	30

Breakdown of Assignments

1. In-class participation

Students are expected to come to class, participate in discussions, and ask questions. Many of the lectures will follow the other assignments so materials covered in class will help students develop their own projects.

2. Annotated bibliographies, develop research questions/hypotheses, and find guiding theory

Students will be expected to read literature on their topic to see what has been published and what can be future directions for the project area. From the literature review, students will be expected to create research questions and/or hypotheses about their project. Additionally, students should have a guiding theory guiding their research project.

3. Advisor check in

Students without advisors will look up potential advisors. This assignment contains several sections, including:

- How to reach out to potential advisors?
- How to use meeting times effectively?
- What to bring to advisor meetings?
- How to communicate with your advisor?

and will end with students getting their research questions and/or hypotheses approved by their advisor.

4. Create a timeline for Data H195A & H195B

Students will develop a weekly timeline of what they will accomplish for the fall and spring semester regarding their project. The timeline will include requirements for the semester, such as reading literature, creating research questions/hypotheses, cleaning data, writing, data documentation. Students will then sign the document and get their advisor's signature to indicate both know what is needed to complete the project.

5. Preliminary proposal

Students will submit their preliminary proposal, including feedback from the instructor from previous assignments. Students will also report reflections on the research process for their projects.

6. Methodology proposal

Students should discuss the methodology proposed for their research project. This information will include:

- How the data was collected
- Details about the dataset
- Details about the sample
- How the methods fit within each student's domain emphasis
- Details about the measure(s) used (e.g., reliability, validity)
- Details about missing data and how it will be handled
- Detailed description about the analytic approach

7. Create a GitHub/MaxQDA/Berkeley Box repo

For students working with quantitative data, the reproducibility workflow will be set up with a GitHub repo, versions of datasets, data documentation (i.e., codebook), and output files. Documentation of code and the research process will be established as a set of best practices to be followed once the research project is underway. This framework for sharing code and results with a project partner or advisor will follow a rubric.

For students working with qualitative data and corresponding analyses, the workflow will include setting up a Berkeley Box account to hold qualitative data (i.e., interview/focus group/etc.) and documentation about organization of participant data, as well as documentation on qualitative coding. This framework for sharing results with a project partner or advisor will follow a rubric.

8. Presenting project in group setting

Students will present and defend their project proposal in a group setting. Peer students will offer feedback and identify clarifications.

9. Final proposal

The culmination of all these efforts will be a paper that summarizes and includes the information from the previous steps, and presents a tight proposal for the research project. Accompanying the paper will be a slide deck for the oral presentation and a github repository for the project work to be managed through. Students will identify a project mentor and/or a project partner where appropriate.

List of Topics

1. Identifying a Research Project
   - Background Literature
   - What has been published
   - Ethics and Human Contexts
   - References
2. Writing an Academic Research Paper
   - Connection to Domain Emphasis
   - Find a journal

    - Structuring an Introduction
    - Develop research question(s)/hypothesis(es)
    - Find a guiding theory/theoretical framework
3. Methodology
    - Get information:
        - Dataset (e.g., procedures)
        - Sample
        - Measures
        - Analytic Approach
4. Project Management
    - Create a GitHub repo or MaxQDA
    - Data Documentation
        - Codebook
4. Thesis Prospectus
    - Connection with faculty advisor
    - Finding an appropriate data source
        - Developing/Further developing documentation

Generative AI Policy

As a matter of general policy, **AI is ALLOWED** in the following contexts:

- Brainstorming and idea generation
    - Students may use AI tools to generate initial ideas or questions
        - Students will document prompts used when using AI tools
- Coding assistance
    - Students may use AI tools to debug code or understand code syntax
        - Students will document prompts for debugging code/understanding code syntax
- Grammar and style checking
    - Students may use AI tools for grammar checks and style checking
        - Students will document prompts for grammar checks and style checking

AI is PROHIBITED in the following contexts:

- Generating sentences, paragraphs, or entire assignments
    - Any and all writing must be completed by the student
- Summarizing any reading, assignment, or instructions
    - Students must read and engage with the course materials yourself

Safe Space Classroom

All students are expected to respect one another in addition to respecting the instructor. This respect includes, but is not limited to, active listening to peers' discussions or questions and openness to all discussion points/questions. Being accepting of other discussion points does not mean that you must agree with those points, but this should lead to respectful discussions and not attacking one another.

Some examples of unacceptable disrespective behaviors:

1. "That is a stupid question."

2. “Are you kidding? We just went over that.”
3. Nonverbal cues of not paying attention or eye rolling

Attendance & Participation

All students are expected to come to every class. Additionally, **GET USED TO TALKING**. This course will be part lecture and part seminar. This means that I will talk through some slides on each week’s topic with points of discussions, but we will also work in small groups, and talking through our project idea. If you cannot make the class, it is the student’s responsibility to email me to let me know they won’t make it to class.

Course Conflicts & Video Recordings

For those students that have course conflicts, I expect you to attend this course. I will record lectures, but I will **not** post them on bCourses unless there are unexpected circumstances or the student and I agree to an arrangement. If a midterm and/or final exam overlaps with this course, it is the student’s responsibility to work out an arrangement with the instructor.

Calendar

Dates	Topic/Speaker/Activity	Readings/Assignments (Due Date at 11:59pm PST)
Week 1		
Week 2 Sept 1	Introductions, syllabus, & JP proposal	
Week 3 Sept 8	Communicating with faculty	Reach out to potential advisor(s)
Week 4 Sept 15	Guest lecturer (Rachel Torres)	Think about data sources Check in on advisor/data
Week 5 Sept 22	Reading literature	Annotated bibliographies (due 10/4)
Week 6 Sept 29	Creating research questions and/or hypotheses Introduction to theory	Creating own research questions and/or hypotheses (due 10/11)
Week 7 Oct 6	Creating a timeline	Get research questions/hypotheses approved (due 11/1) Create a timeline for fall & spring semesters (due 11/1)
Week 8 Oct 13	Introduction to academic writing	Check in on advisor/data Start drafting an outline for final proposal
Week 9 Oct 20	Methods	Methodology proposal (due 11/8)
Week 10 Oct 27	Presenting to different audiences	
Week 11 Nov 3	Preliminary proposal pitches	Preliminary proposal (due 11/2)
Week 12 Nov 10	GitHub & reproducibility	Check in on advisor/data Create repo (due 11/22)
Week 13 Nov 17	Student Presentations	
Week 14 Nov 24	Check ins through Zoom	
Week 15 Dec 1	Student Presentations	
Week 16 Dec 8	No class	Final proposals (due 12/13)